

9 (a) Not affected by external factors / conditions [B1]

(b) (i) Total mass of products = $218.0090 + 4.0026 = 222.0116 \text{ u}$

Total mass of reactant = 222.0176 u

Difference in total mass = 0.0060 u [M1]

Conversion from u to kg = $00060 \times 1.66 \times 10^{27} = 9.96 \times 10^{30} \text{ kg}$

Energy liberated = $mc^2 = (9.96 \times 10^{30}) \times (3.0 \times 10^8)^2 = [\text{C1}]$
 $= 8.96 \times 10^{-13} \text{ J}$ [A1]

(ii) Since total momentum is conserved, and initial momentum of Rn-222 was zero,

The total momentum of the decay products must add up to zero. [M1]

The Po-218 and alpha particle move in opposite directions with equal (magnitude of) momentum. [B1]

Since Po-218 has a larger mass than the alpha particle, its velocity will be lower.

(iii) Since the mass of alpha particle / mass of Po-218 = $4 / 218$

The speed of alpha particle / speed of Po-218 = $218 / 4$ [B1]

Hence the kinetic energy of alpha particle / kinetic energy of Po-218 = $218/4 = 54.5$
[A0 (Using exact values given, 54.47)]

(c) (i)

	Number of neutrons	Number of protons	Mass of constituent nucleons in u	Mass defect in u	<u>mass defect</u> <u>number of nucleons</u>
$^{222}_{86}\text{Rn}$	$222 - 86 = 136$	86	$(86 \times 1.00727) + (136 \times 1.00866) = 223.80298$	$223.80298 - 222.0176 = 1.78538$	$1.78538 / 222 = 8.04225 \times 10^{-3}$

[C1 for correct number of neutrons and protons for both Rn-222 or Po-218]

[C1 for correct substitution / value for mass defect for Rn-222 and Po-218]

[A1 for correct final answers]

(ii) Since mass defect is directly proportional to binding energy, a higher value of mass defect per nucleon means a higher binding energy per nucleon. [B1]

On average, it requires higher energy to break apart Po-218 into its constituent nucleons, compared to Rn-222. [B1]

Hence, Po-218 is more stable. [A1]

(d) (i) It is the time taken for half the original number of radioactive nuclei to decay. [B1]

(ii) Decay constant = $\ln 2 / t_{1/2} = \ln 2 / (3.82 \times 24 \times 60 \times 60) = 0.00000210014$

(iii) Not valid [A0] as activity also depends on amount of radioactive material present [M1], and not just the half-life. ($A = \lambda N$)

(e) The observation suggests that Po-218 is unstable [B1]

The additional alpha particles detected come from the decay of Po-218 (and its daughter nuclides). [B1]

The almost-immediate decay of Po-218 (and its daughter nuclides) after it is produced shows that Po-218 has a very short half-life (half-lives), i.e. further alpha decays [B1]